$\mathbf{EFX_0}$ Allocations with at Most Three Relevant Goods

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Abstract. An allocation of indivisible goods is EFX_0 if no agent envies another agent's bundle after removing any single good from it, even one the envious agent values at zero. We prove that every instance with nonnegative additive valuations in which each agent positively values at most three goods has a complete EFX_0 allocation in which all bundles but at most one have at most two goods, and an algorithm, Draft and Exchange (DE), that computes one in time $O(n(n+m))$ with at most $4n$ trades. After a draft, goods are traded along a chain or a ring of agents, each of which gets a higher-ranked holding and loses no value, until a free agent can take all the leftover goods (or every agent holds its pair). The key fact is that when no free agent can take them, every agent can point at an agent that would gladly take its good, and the pointers close a ring. The proof is machine-checked in Lean, together with the algorithm, its bound of $4n$ trades and the running time of an implementation.

Keywords: Fair division · Indivisible goods · EFX · Trading cycles

1 Introduction

Agents N ($|N| = n \geq 1$) share a set M of m indivisible goods with nonnegative additive valuations $v_i(S) = \sum_{g \in S} v_i(g)$. An allocation X assigns every good to exactly one agent, who receives the bundle X_i; no good is left out (allocations are *complete*). It is *EFX*$_0$ if

$$v_i(X_i) \;\geq\; v_i(X_j \setminus \{h\}) \qquad \text{for all } i \neq j \text{ and all } h \in X_j,$$

and *EFX* asks this only for goods h with $v_i(h) > 0$ [3]. So EFX_0 implies EFX, and the two coincide for positive values. Under EFX_0, if X_j contains a good that i values at 0, then i must not envy X_j at all. Whether complete EFX allocations always exist for additive valuations is open; they exist in special cases [5,7,12,2,4,1,6,13,8,9] (Table 1 in Appendix E). Let $R_i = \{g : v_i(g) > 0\}$, the *relevant* goods of i. If $|R_i| \leq 2$ for all i, serial dictatorship gives EFX_0 [11]. At three goods this breaks: an agent that values a, b, c at 4, 3, 2 may take a and then see b and c end up in the last agent's bundle with a third good; removing that third good leaves b and c, worth $5 > 4$.

Theorem 1 (existence and shape). *Every instance with $n \geq 1$ agents and nonnegative additive valuations in which every agent positively values at most three goods has a complete EFX$_0$ allocation in which all bundles but at most one have at most two goods.*

Theorem 2 (algorithm). *Algorithm* DE *(Section 6) computes such an allocation with at most n peeling steps and at most $4n$ trades, in $O(n(n+m))$ steps of a unit-cost random-access machine. Every decision of* DE *compares two sums of at most two values of one agent, and each such comparison counts as one step.*

What is new. As far as we know,[1] no earlier result settles the case of three goods. Viswanathan and Mehta [14] study the same restriction, up to four goods per agent, but for ordinary EFX and with only a proof sketch. Their result does not imply ours, because EFX does not give EFX_0, and the difference matters exactly when agents value few goods, so that most goods are worth 0 to most agents. In the example above, let the agent hold a, worth 4, and let another bundle be $\{b, c, g\}$ with $v(g) = 0$. Removing b or c leaves value 2 or 3, so EFX holds; removing g leaves $5 > 4$, so EFX_0 fails. (Their algorithm gives the goods nobody values to an arbitrary agent, which can create such a bundle.) Among the existence results for complete EFX_0 allocations surveyed in Table 1, none covers all instances with at most three relevant goods per agent: each bounds the number of agents, the number of goods compared with agents, the number of agents valuing a good, the girth of the hypergraph of goods, or the number of distinct valuations, none of which our model bounds, or allows at most two relevant goods per agent [11]. Appendix E gives an instance with six agents that none of them covers, and checks it row by row. We contribute the existence (Theorem 1); the shape, which has no counterpart we know of and cannot be strengthened in two natural ways: some instances have no EFX_0 allocation with all bundles of at most two goods, and on an instance with four agents every EFX_0 allocation has bundle sizes $4, 1, 1, 1$, so the large bundle cannot be limited to three goods (Appendix D); the algorithm (Theorem 2); and a machine-checked proof (Appendix F lists what exactly is checked).

How the proof works. Agents that can leave with at most one good do so; the others draft at most one good each. Then, until a *free* agent can take all the leftover goods (or every agent holds its pair), goods are traded

[1] Literature search of 25 September 2026 (arXiv, the web, Zenodo, Google Scholar, Crossref, the EC 2026 programme), recorded in `proofs/novelty.md` of [10].

along a *chain* or a *ring* of agents, each of which gets a higher-ranked holding (its pair ranks above its top) and loses no value. Termination is measured by the total score, the sum of these ranks, not by welfare: an agent moving between two goods of equal value gains score but no value. Section 6 ends with a small example.

2 Peeling

The proofs of Sections 2 and 3 are in Appendix A.

Agents and goods are numbered, and "index order" and "the first" refer to these numbers. A good is *alone* in an allocation if it is the only good of its bundle.

Lemma 1 (peeling). *Let $i \in N$ and $P \subseteq M$ with (i) $v_i(P) \geq v_i(M \setminus P)$ and (ii) $|P| \leq 1$. If the instance with agents $N \setminus \{i\}$ and goods $M \setminus P$ has an EFX$_0$ allocation Z, then Z together with $X_i = P$ is an EFX$_0$ allocation of the whole instance.*

Rule R1. Let G be the set of remaining goods, and p agent i's favourite good of G (largest value; between equal values, the lower index). If $G = \emptyset$ or $v_i(p) = 0$, so that i values no remaining good, then i may leave with $P = \emptyset$. If $v_i(p) > 0$ and $v_i(G \setminus \{p\}) \leq v_i(p)$, then i may leave with $P = \{p\}$. Rule R1 applies, for instance, to every agent that values at most two remaining goods; so if every agent values at most two goods, serial dictatorship is EFX$_0$ [11].
When $|R_i| = 3$ we name agent i's relevant goods a_i, b_i, c_i so that $v_i(a_i) \geq v_i(b_i) \geq v_i(c_i) > 0$, breaking ties in favour of the lower index. This gives a strict *ranking* $a_i \succ_i b_i \succ_i c_i$, and $g \succ_i h$ implies $v_i(g) \geq v_i(h)$. Agent i is *strictly balanced* if $v_i(a_i) < v_i(b_i) + v_i(c_i)$.

Lemma 2 (when nobody can leave). *Let agent i value at most three goods of the set G of remaining goods, and suppose that rule R1 does not apply to i. Then i values exactly three goods of G and is strictly balanced with respect to them.*

In Sections 3–5 we are in the *core case*: every agent values exactly three goods and is strictly balanced. Goods that no agent values may exist.

3 States

Definition 1 (states and scores). *A* state *Y gives each agent i a* holding *Y_i: nothing, $\{a_i\}$, $\{b_i\}$, $\{c_i\}$, or its* pair *$\{b_i, c_i\}$, with the holdings pairwise disjoint. U is the set of agents holding their pair. The goods held by*

nobody are leftover*; J is the set of leftover goods. Agent i's* score *is* 4, 3, 2, 1 *or* 0 *if it holds its pair, a_i, b_i, c_i or nothing. Since $v_i(a_i) < v_i(b_i)+v_i(c_i)$, a higher score never means a lower value.*

A higher score need not mean a higher value: if $v_i(a_i) = v_i(b_i)$, moving from b_i to a_i raises the score but not the value. Gaining the pair, or a good after holding nothing, does raise the value. The trades below raise scores; the total score $\Phi(Y) = \sum_i \text{score}_i(Y)$, an integer between 0 and $4n$, measures progress.

Definition 2 (wants, validity, free agents). *Let Y be a state. An agent $i \notin U$* wants *a good $g \in R_i$ if it holds nothing, or holds a good y with $g \succ_i y$. Agents in U want nothing. Y is* valid *if every good that some agent wants is held alone: it is the whole holding of some agent. An agent $i \notin U$ is* free *if it holds nothing or holds a good that nobody wants. F is the set of free agents.*

No agent wants its own good, so in a valid state the holder of a wanted good is another agent, outside U and not free. Leftover goods and the goods of pair holders are not wanted in a valid state. An agent outside $U \cup F$ holds a single good that some agent wants.

Validity is what makes finishing possible. It keeps every wanted good alone, so a bundle of two or more goods may contain only goods that nobody wants. An agent holding nothing wants all its goods, and an agent holding a single good wants every good it ranks higher; so such a bundle contains none of the first agent's goods, and for the second only goods worth at most its own. One obstruction remains, and only with three goods: an agent holding its top a_x may find b_x and c_x together in a bundle with a third good, and $v_x(b_x) + v_x(c_x) > v_x(a_x)$. Section 4 deals with exactly this case.

Lemma 3 (the draft). *Let the agents, in any order, each take the first of a_i, b_i, c_i not yet taken, or nothing if all three are taken. The result is a valid state with $U = \emptyset$.*

Lemma 4 (staying valid). *Let Y be valid, and Y' a state in which every agent's score is at least its score in Y and every good wanted in Y is held alone. Then Y' is valid.*

4 Finishing

Definition 3 (blockers, finishing). *Let Y be a state and o a free agent. A* blocker *of o is an agent $x \neq o$ that holds exactly its top, $Y_x = \{a_x\}$, and*

has $b_x, c_x \in J \cup Y_o$. Agent o can finish *with a set $H \subseteq J$ if $|H| \leq |F| - 1$ and H contains b_x or c_x for every blocker x of o. Its* completion *X gives each good of H to a different free agent other than o, gives o the bundle $X_o = Y_o \cup (J \setminus H)$, and lets every other agent keep its holding (plus its good of H, if any).*

So a blocker of o is an agent that would find both b_x and c_x in o's bundle if o took all the leftover goods; H moves one of the two out of it. When every agent holds its pair, the completion with any agent o gives o the bundle $Y_o \cup J$ and lets the others keep their pairs.

Theorem 3 (soundness). *Let Y be a valid state, and X the completion with a free agent o that can finish with a set H, or, if every agent holds its pair, with any agent o. Then X is an EFX$_0$ allocation, and every bundle other than X_o has at most two goods.*

Proof. Sizes. A free agent other than o holds at most one good and receives at most one good of H; a pair holder holds two goods; every other agent keeps a single good.

Wanted goods stay alone. Let g be wanted, held alone by j. Then j is not free, so $j \neq o$ and j receives no good of H: $X_j = \{g\}$. Hence every good in a bundle of two or more goods is unwanted.

No envy. Let $i \neq j$ and $h \in X_j$. We show $v_i(X_j \setminus \{h\}) \leq v_i(X_i)$, where $v_i(X_i) \geq v_i(Y_i)$. If $|X_j| \leq 1$, then $X_j \setminus \{h\} = \emptyset$. Let $|X_j| \geq 2$, and $T = X_j \cap R_i$; it contains no good that i wants and none of i's holding.

- If i holds its pair, then $T \subseteq \{a_i\}$, and $v_i(T) \leq v_i(a_i) < v_i(b_i) + v_i(c_i)$.
- If i holds nothing, it wants all of R_i, so $T = \emptyset$.
- If $Y_i = \{y\}$, every good of T is ranked below y and worth at most $v_i(y)$. If $|T| \leq 1$, then $v_i(X_j \setminus \{h\}) \leq v_i(T) \leq v_i(y)$. If $|T| = 2$, then $y = a_i$ and $T = \{b_i, c_i\}$. If $X_j = \{b_i, c_i\}$, removing h leaves one good, worth at most $v_i(b_i) \leq v_i(a_i)$. Otherwise $|X_j| \geq 3$, so $j = o$ by the sizes, and $b_i, c_i \in X_o \subseteq Y_o \cup J$: i is a blocker of o. Then H contains b_i or c_i, which went to an agent other than o, a contradiction. (When every agent holds its pair, only the first case occurs.) □

5 Improving

Definition 4 (rings). *Let Y be a state. A* ring *is a cycle $w_0 \to w_1 \to \cdots \to w_{k-1} \to w_0$ of $k \geq 2$ distinct agents outside U, each holding a single good $y_t = y_{w_t}$, such that each arrow $w_t \to w_{t+1}$ (indices modulo k) is*

- *a* want arrow*: w_{t+1} wants y_t; or*
- *a* pair arrow*: w_t is free, w_{t+1} is a blocker of w_t, and the pair of w_{t+1} is $\{y_t, h_{t+1}\}$ with $h_{t+1} \in J$;*

and the leftover goods h_{t+1} of the pair arrows are pairwise distinct. Trading along the ring *gives each w_{t+1} the good y_t, together with h_{t+1} for a pair arrow, in place of its own good; agents off the ring keep their holdings.*

Arrows point the way goods move, not from an agent to the holder of a good it wants: along $w_t \to w_{t+1}$, the good of w_t goes to w_{t+1}. In a valid state, an arrow from a free agent is a pair arrow (nobody wants its good), and an arrow from any other agent is a want arrow.

Lemma 5 (ring). *Trading along a ring of a valid state gives a valid state in which every agent of the ring has a higher score and every other agent the same score.*

Proof. A state. Each good y_t moves to w_{t+1} only, and the goods h_{t+1} are distinct leftover goods; agents off the ring keep goods that are neither. The new holdings are a single good of the receiver or its pair. *Scores.* Along a want arrow, w_{t+1} receives a good it ranks above what it held. Along a pair arrow, w_{t+1} goes from its top to its pair. *Wanted goods.* Let g be wanted in Y, held alone by j. If j is off the ring, it keeps g. If $j = w_t$, then w_t is not free, so its arrow is a want arrow, and w_{t+1} now holds g alone. Lemma 4 gives validity. □

Lemma 6 (chain). *Let Y be valid, and x an agent with $Y_x = \{a_x\}$ and $b_x, c_x \in J$. Let $j_0 = x$, and as long as j_t holds a good that some agent wants, let j_{t+1} be the first such agent. If an agent repeats, the agents from its first occurrence on form a ring of want arrows. Otherwise the sequence stops at an agent j_k ($k \geq 0$) that holds nothing or a good nobody wants, and the following gives a valid state in which $x, j_1, \ldots, j_k$ have higher scores and every other agent the same score: x takes its pair; each j_t with $1 \leq t \leq k$ takes the good of j_{t-1}; the old good of j_k, if any, becomes leftover.*

Proof. Agent x holds its top and wants nothing, so $j_t \neq x$ for $t \geq 1$. If $j_s = j_t$ with $1 \leq s < t$ is the first repeat, then $j_s, \ldots, j_{t-1}$ are distinct agents outside U, each holding a good that the next one wants: a ring of want arrows ($t - s \geq 2$, as no agent wants its own good). Otherwise: the goods b_x, c_x were leftover, and each moved good moves one step, so this is a state. Agent x goes from its top to its pair, and each j_t, $t \geq 1$,

receives a good it wants. A wanted good is not b_x or c_x (they are leftover); if its holder is off the path it stays, and if its holder is j_t, then $t < k$ (the good of j_k is not wanted), and j_{t+1} now holds it alone. Lemma 4 gives validity. □

Applying a chain, or trading along a ring, is a *trade*. By Lemmas 5 and 6, a trade keeps the state valid and raises the total score Φ by at least one. From now on, assume that no chain applies:

(NC) no agent x with $Y_x = \{a_x\}$ has both b_x and c_x in J.

Lemma 7 (the finishing test). *Let Y be a state with (NC), and o a free agent.*

(a) If o holds nothing, it has no blocker.
(b) If x is a blocker of o, then o holds a good y_o, and the pair of x is $\{y_o, h_x\}$ for a leftover good h_x.
(c) Let H_o be the set of the goods h_x of the blockers x of o. Then o can finish if and only if $|H_o| \le |F| - 1$, and then it can finish with H_o. In particular a free agent holding nothing can finish.

Proof. Let x be a blocker of o: $b_x, c_x \in J \cup Y_o$, and not both are in J, by (NC). So one of them is in Y_o, which proves (a), and, as o holds at most one good, that good is y_o and the other one, h_x, is leftover: (b). A set $H \subseteq J$ contains b_x or c_x if and only if it contains h_x; so the sets H allowed by Definition 3 are those with $H_o \subseteq H \subseteq J$ and $|H| \le |F| - 1$, which exist if and only if $|H_o| \le |F| - 1$. If o holds nothing, $H_o = \emptyset$ and $|F| \ge 1$. □

Theorem 4 (Improvement Lemma). *Let Y be a valid state in which not every agent holds its pair and no free agent can finish. Then a chain or a ring gives a valid state in which every agent has at least its score and some agent a higher one.*

Proof. If (NC) fails, Lemma 6 applies. Assume (NC). By Lemma 7, every free agent o holds a good and has $|H_o| \ge |F|$.

Distinct leftover goods. List the free agents in index order, $o_1, \ldots, o_f$. For $i = 1, \ldots, f$ in turn, let x_{o_i} be the first blocker of o_i whose leftover good is not among $q_{o_1}, \ldots, q_{o_{i-1}}$, and $q_{o_i} = h_{x_{o_i}}$. It exists because the blockers of o_i have $|H_{o_i}| \ge f > i - 1$ different leftover goods.

Pointers. For every agent $j \notin U$ let $\sigma(j) = x_j$ if j is free, and otherwise let $\sigma(j)$ be the first agent that wants the good of j (one exists, as $j \notin U \cup F$). In both cases $\sigma(j) \notin U$ and $\sigma(j) \ne j$.

The ring. Start at the first agent outside U and follow σ until an agent repeats. The agents from its first occurrence on form a cycle $w_0 \to \cdots \to w_{k-1} \to w_0$ with $k \geq 2$, each holding a single good (free agents by Lemma 7). The arrow from a free agent w_t is a pair arrow, with $h_{t+1} = q_{w_t}$, by Lemma 7(b); these goods are distinct for distinct free agents. Every other arrow is a want arrow. So the cycle is a ring, and Lemma 5 applies. □

6 The Algorithm Draft and Exchange

Algorithm 1 lists DE; ties are broken by index as in Section 2.

Theorem 5 (DE is correct). *On every instance in which every agent values at most three goods,* DE *returns a complete EFX$_0$ allocation in which all bundles but at most one have at most two goods. It peels at most n agents and makes at most $4n$ trades.*

Proof. Peeling. Each round removes one agent, so there are at most n rounds. Rule R1 for the agent i peeled with P_i is conditions (i) and (ii) of Lemma 1 on the instance left before it, so by induction from the last

Algorithm 1 Draft and Exchange (DE)

Require: agents N, goods M, nonnegative values $v_i(g)$, every agent with $|R_i| \leq 3$
$A \leftarrow N$, $G \leftarrow M$
while $|A| \geq 2$ and rule R1 applies to an agent of A, for G **do** ▷ peeling
  $i \leftarrow$ the first such agent; $X_i \leftarrow P_i$ (by rule R1); $A \leftarrow A \setminus \{i\}$, $G \leftarrow G \setminus P_i$
if $|A| = 1$ **then** give G to the agent of A and **return** X
for $i \in A$ in index order **do** ▷ the draft
  $Y_i \leftarrow$ the first of $\{a_i\}, \{b_i\}, \{c_i\}$ whose good is not taken, else nothing
loop
  if some x with $Y_x = \{a_x\}$ has $b_x, c_x \in J$ **then** ▷ chain
    apply the chain of the first such x, or the ring of want arrows it runs into (Lemma 6)
  else if every agent holds its pair **then** ▷ finish
    $o \leftarrow$ the first agent; $H \leftarrow \emptyset$; **break**
  else if some free agent o has $|H_o| \leq |F| - 1$ **then** ▷ finish
    $o \leftarrow$ the first such agent; $H \leftarrow H_o$; **break**
  else ▷ ring
    trade along the ring of the proof of Theorem 4
give the k-th good of H (in the order of first blockers) to the k-th free agent other than o
$X_o \leftarrow Y_o \cup (J \setminus H)$; every other $i \in A$ keeps Y_i (plus its good of H, if any)
return X

round down it suffices that DE returns an EFX$_0$ allocation of the agents A and goods G left at the end. If $|A| = 1$, the single agent faces no other bundle.

The core. Let $|A| \geq 2$. Rule R1 applies to no agent of A, so by Lemma 2 this is the core case. The draft gives a valid state (Lemma 3). A chain or a ring gives a valid state in which nobody's score drops and some score rises (Lemmas 6 and 5; the ring exists when it is used, by the proof of Theorem 4). The total score Φ is an integer between 0 and $4|A|$ and rises by at least one with every trade, so there are at most $4|A| \leq 4n$ trades, and the loop stops. When it stops, o can finish with H (Lemma 7(c); the chain test failed, so (NC) holds), or every agent holds its pair; Theorem 3 gives an EFX$_0$ allocation in which only X_o can have more than two goods. Peeled agents receive at most one good. □

A small example. Agents z, x, o, in index order, rank $g_4 \succ g_0 \succ g_1$, $g_0 \succ g_1 \succ g_2$ and $g_0 \succ g_4 \succ g_1$, with values 4, 3, 2; nobody values g_3. Nobody is peeled. In the draft z takes g_4, x takes g_0, and o takes g_1; the leftover goods are g_2, g_3. Agent o wants g_0 and g_4, which are held alone: the state is valid, and o is the only free agent. No chain applies. Agent x is a blocker of o: if o took $\{g_1, g_2, g_3\}$, x would remove g_3 and see g_1 and g_2, worth $5 > 4$. So $H_o = \{g_2\}$, there is no other free agent to take g_2, and o cannot finish. The pointers are $\sigma(o) = x$, a pair arrow (x's pair is o's good g_1 and the leftover g_2), and $\sigma(x) = \sigma(z) = o$, want arrows ($o$ wants g_0 and g_4). Following them from z closes the ring $o \to x \to o$: x takes $\{g_1, g_2\}$ and o takes g_0. Now nobody wants anything, the free agents are z and o, and z has no blocker, so z finishes: $X_z = \{g_4, g_3\}$, $X_x = \{g_1, g_2\}$, $X_o = \{g_0\}$, an EFX$_0$ allocation.

Running time. We count steps of a unit-cost random-access machine: an array access, an arithmetic operation on indices, and a comparison of two sums of at most two values of one agent each take one step. This is not a bound on bit complexity for arbitrary encodings of the values. DE keeps an array of the holder of each good (or a mark that it is leftover or peeled) and an array of the holding of each agent.

- *Input.* One pass over the nm values finds each agent's at most three relevant goods, and sorting them gives its ranking: $O(nm)$ steps.
- *Peeling.* Rule R1 for agent i looks only at its relevant goods still in G: it finds its favourite p and compares $v_i(p)$ with the sum of the at most two other values. So a round tests every agent in $O(n)$ steps and removes at most one good.

– *A loop iteration* takes $O(n+m)$ steps. Each agent outside U marks the at most three goods it wants, and each good records the first agent, in index order, that wants it; then the leftover goods, the free agents, and the chain test (look up the holders of b_x and c_x for every x holding a_x) take one pass each. A chain follows the recorded first wanting agents, with marks on visited agents, in $O(n)$ steps. Under (NC), an agent x holding a_x is a blocker of o exactly when one of b_x, c_x is leftover and the other is the good of the free agent o (Lemma 7). So each agent is a blocker of at most one free agent, and one pass in index order lists the blockers of every free agent, each list in index order. One pass over these lists, with marks on goods, gives the sizes $|H_o|$, and another the distinct leftover goods q_o of the proof of Theorem 4. Then each $\sigma(j)$ takes one step, and following σ with marks on visited agents finds the ring in $O(n)$ steps. A trade, and the final completion, take $O(n+m)$ steps.

With at most n peeling rounds and at most $4n+1$ loop iterations (the last one finishes), DE takes $O(nm+n^2+n(n+m)) = O(n(n+m))$ steps. Lean proves a bound of $750(n+1)(n+m+1)$ steps for a counted program (Appendix F).

Real values. Every decision of DE compares a value of an agent with another or with the sum of two of its values, and the proofs use nothing else about the values. So Theorem 5 and these remarks prove Theorems 1 and 2, also for real values. Appendix F explains how the Lean proof reaches real values.

7 Further Results, Verification and Open Problems

Appendix B has a larger example, where DE's ring gives pairs to two agents at once; rings may need any number of pair arrows (Appendix C), and some instances need a bundle of three or four goods (Appendix D). Every result here is machine-checked in core Lean 4 (`lean/EFX/K3DE*.lean` in [10]): the main theorem for natural-number values, real values through natural numbers that answer the same comparisons, and the running time for a counted program that returns DE's output; Appendix F maps each result to its Lean theorem (Table 2) and says what is not formalized. Tests found no failure; the reviews were by AI agents, with no human peer review yet. Open problems: Does every instance in which each agent positively values at most four goods have a complete EFX_0 allocation? (For ordinary EFX, Viswanathan and Mehta [14] announce

one.) Do fewer than $4n$ trades suffice? What is the least bound on the size of the large bundle that holds for every instance?

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References

1. Afshinmehr, M., Ashuri, A., Mahmoudkhan, P., Mehlhorn, K., Shahrezaei, A.M.: EFX allocations exist on multi-graphs. arXiv preprint arXiv:2606.18665 (2026), accepted at the 27th ACM Conference on Economics and Computation (EC 2026)
2. Alkassar, E., Fouz, M., Mehlhorn, K.: Complete EFX allocations exist for four additive agents and up to nine goods. arXiv preprint arXiv:2608.08590 (2026)
3. Amanatidis, G., Birmpas, G., Filos-Ratsikas, A., Voudouris, A.A.: Fair division of indivisible goods: A survey. In: Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22). pp. 5385–5393 (2022). `https://doi.org/10.24963/ijcai.2022/756`, arXiv:2202.07551
4. Bratby, J.: Complete EFX allocations with four more goods than agents. Zenodo preprint, version 1, `https://doi.org/10.5281/zenodo.22665180`; claims complete EFX_0 allocations when $m = n + 4$; not peer-reviewed (2026)
5. Chaudhury, B.R., Garg, J., Mehlhorn, K.: EFX exists for three agents. Journal of the ACM **71**(1), 4:1–4:27 (2024). `https://doi.org/10.1145/3616009`, conference version in Proc. 21st ACM Conference on Economics and Computation (EC 2020), pp. 1–19; full version arXiv:2002.05119
6. Christodoulou, G., Fiat, A., Koutsoupias, E., Sgouritsa, A.: Envy-freeness up to any good allocations on graphs. Mathematics of Operations Research (2026). `https://doi.org/10.1287/moor.2025.0915`, conference version: "Fair Allocation in Graphs", in Proc. 24th ACM Conference on Economics and Computation (EC 2023), pp. 473–488
7. HV, V.P., Ghosal, P., Nimbhorkar, P., Varma, N.: EFX exists for three types of agents. In: Proceedings of the 26th ACM Conference on Economics and Computation (EC 2025). pp. 101–128. ACM (2025). `https://doi.org/10.1145/3736252.3742509`, arXiv:2410.13580
8. Kaviani, A., Seddighin, M., Shahrezaei, A.: Almost envy-free allocation of indivisible goods: A tale of two valuations. In: Web and Internet Economics (WINE 2024). pp. 261–276. Lecture Notes in Computer Science, Springer (2026). `https://doi.org/10.1007/978-3-032-08560-3_15`, full version arXiv:2407.05139
9. Lianeas, T., Sgouritsa, A., Sotiriou, M.M.: EFX allocation in (multi)hypergraphs. Proceedings of the AAAI Conference on Artificial Intelligence **40**(20), 17102–17110 (2026). `https://doi.org/10.1609/aaai.v40i20.38759`, AAAI-26; arXiv:2608.03171
10. Lin, E.: efx0-three-goods-proof: EFX_0 with at most three relevant goods per agent (paper, Lean formalization, tests, referee reports). GitHub repository, tag `v1.0`, `https://github.com/evanlin23/efx0-three-goods-proof` (2026)
11. Lin, E., Osterhaus, G.: Strong EFX allocations for 2-relevant agents: Lean 4 formalization. GitHub repository `mrd-efx`, tag `v1.2.0` (commit `fa42aa1`, 23 September 2026), `https://github.com/evanlin23/mrd-efx` (2026)

12. Mahara, R.: Extension of additive valuations to general valuations on the existence of EFX. Mathematics of Operations Research **49**(2), 1263–1277 (2024). https://doi.org/10.1287/moor.2022.0044, conference version in Proc. 29th Annual European Symposium on Algorithms (ESA 2021), LIPIcs 204, 66:1–66:15; arXiv:2107.09901
13. Sgouritsa, A., Sotiriou, M.M.: On the existence of EFX allocations in multigraphs. arXiv preprint arXiv:2502.09777 (2025), extended abstract, without the bipartite case, in Proc. 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025), pp. 2735–2737
14. Viswanathan, M., Mehta, R.: On the existence of EFX under picky or non-differentiative agents. In: Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024). pp. 2534–2536 (2024). https://doi.org/10.65109/osos1146, extended abstract

A Proofs of Sections 2 and 3, and a Corollary

Proof (of Lemma 1). Agent i and a bundle Z_j: $v_i(Z_j \setminus \{h\}) \le v_i(Z_j) \le v_i(M \setminus P) \le v_i(P)$. Agent $j \neq i$ and the bundle P: removing its only good leaves nothing. Agent $j \neq i$ and a bundle Z_k: this is a condition of Z. □

Proof (of Lemma 2). Rule R1 fails, so $v_i(p) > 0$ and $v_i(G \setminus \{p\}) > v_i(p)$. If i valued at most two goods of G, then $v_i(G \setminus \{p\})$ would be the value of at most one good, at most $v_i(p)$. So i values exactly three goods of G, p is its top, and $v_i(b_i) + v_i(c_i) > v_i(a_i)$. □

Proof (of Lemma 3). Each agent holds one of its goods or nothing, and no good is taken twice. If i wants g, then g was taken before i's turn, by an earlier agent as its only good, and holdings never change during the draft. □

Proof (of Lemma 4). Let i want g in Y'. If i holds nothing in Y', its score in Y was 0, so it held nothing in Y and wanted g there too. If i holds a single good y' in Y', its score in Y was below 4, so in Y it held nothing, or a good y with $y' = y$ or $y' \succ_i y$; since $g \succ_i y'$, again i wanted g in Y. So g is held alone in Y'. □

Corollary 1 (score-Pareto-maximal valid states). *In the core case, in every valid state that no valid state Pareto-dominates in scores, either every agent holds its pair or some free agent can finish. In particular every core instance has a complete EFX_0 allocation in which all bundles but at most one have at most two goods.*

Proof. The first part is Theorem 4. Valid states exist (Lemma 3), there are finitely many, and one with the largest total score is not Pareto-dominated in scores; Theorem 3 finishes. □

This is a second existence argument, within the valid states; it is not an efficiency guarantee. Maximality is in scores and among valid states only, and nothing is claimed about the efficiency of the resulting allocation, or of DE's.

B A Worked Example

On this instance every valid state that Pareto-dominates the draft state in scores gives pairs to two agents at once, so a ring with a single pair arrow is not enough (Appendix C). Ten goods $g_0, \ldots, g_9$ and six agents, in index order x_1, x'_1, x_2, x'_2, o_1, o_2, with values 4, 3, 2 on their goods $a \succ b \succ c$ (any strictly balanced values with these rankings give the same run):

agent	$a \succ b \succ c$	after the draft	after the trade
x_1	$g_0 \succ g_4 \succ g_6$	$\{g_0\}$ (a)	$\{g_4, g_6\}$ (pair)
x'_1	$g_1 \succ g_4 \succ g_7$	$\{g_1\}$ (a)	$\{g_1\}$ (a)
x_2	$g_2 \succ g_5 \succ g_8$	$\{g_2\}$ (a)	$\{g_5, g_8\}$ (pair)
x'_2	$g_3 \succ g_5 \succ g_9$	$\{g_3\}$ (a)	$\{g_3\}$ (a)
o_1	$g_2 \succ g_3 \succ g_4$	$\{g_4\}$ (c)	$\{g_2\}$ (a)
o_2	$g_0 \succ g_1 \succ g_5$	$\{g_5\}$ (c)	$\{g_0\}$ (a)

The draft. Nobody is peeled. The four x's take their tops; o_1 finds g_2 and g_3 taken and takes g_4; o_2 takes g_5. Agent o_1 wants g_2 and g_3, o_2 wants g_0 and g_1, and the x's want nothing; the four wanted goods are held alone. The leftover goods are $g_6, \ldots, g_9$, and the free agents are o_1 and o_2. No chain applies: the b of each x is held by o_1 or o_2.

Nobody can finish. The blockers of o_1 are x_1 (pair $\{g_4, g_6\}$) and x'_1 (pair $\{g_4, g_7\}$), so $H_{o_1} = \{g_6, g_7\}$; symmetrically $H_{o_2} = \{g_8, g_9\}$. Both have $|H_o| = 2 > |F| - 1$. Concretely, if o_1 takes the leftovers, only o_2 can take one of g_6, g_7; with $H = \{g_6\}$, $X_{o_1} = \{g_4, g_7, g_8, g_9\}$, and x'_1, holding g_1 (worth 4), values X_{o_1} without g_8 at $3 + 2 = 5$. All ten candidate completions (finisher o_1 or o_2, with H empty or one leftover good) fail the EFX$_0$ test.

No small trade helps. Every want arrow ends at o_1 or o_2, so a ring with no pair arrow would have to leave one of them by a want arrow; but nobody wants their goods. A ring with exactly one pair arrow, from o_1 say, would return to o_1 by want arrows; but x_1 and x'_1 have want arrows only to o_2,

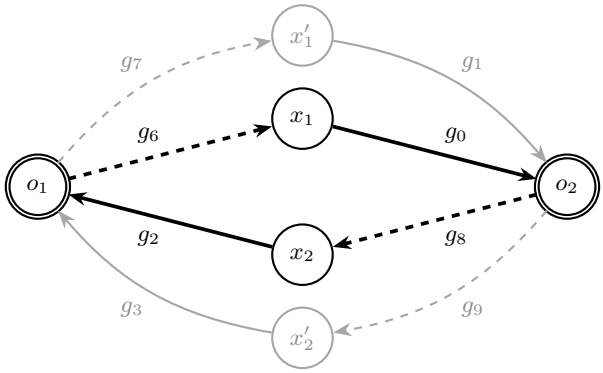


Fig. 1. The arrows of the draft state of Appendix B. Double circles are the free agents. Dashed arrows are pair arrows, labelled by the blocker's leftover good; solid arrows are want arrows, labelled by the good that moves. The ring chosen by DE is drawn in black. Every cycle of these arrows has four agents and two pair arrows.

and the arrows from o_2 are pair arrows. In fact exactly four valid states Pareto-dominate the draft state in scores, and each gives pairs to two agents.

The trade. DE takes the first blocker of each free agent: $x_{o_1} = x_1$, with $q_{o_1} = g_6$, and $x_{o_2} = x_2$, with $q_{o_2} = g_8$. So $\sigma(o_1) = x_1$, $\sigma(x_1) = o_2$ (o_2 wants g_0), $\sigma(o_2) = x_2$ and $\sigma(x_2) = o_1$ (o_1 wants g_2): the ring of Figure 1. Along it, x_1 takes $\{g_4, g_6\}$, o_2 takes g_0, x_2 takes $\{g_5, g_8\}$ and o_1 takes g_2. The total score rises from 14 to 20.

Finishing. Now nobody wants anything, the leftover goods are g_7 and g_9, and the free agents are x'_1, x'_2, o_1, o_2. No chain applies (b of x'_1 is g_4, now held by x_1; similarly for the others). The first free agent, x'_1, has no blocker: no other agent holding its top has both b and c in $\{g_1, g_7, g_9\}$. So x'_1 finishes with $H = \emptyset$, and DE returns

$$X_{x_1} = \{g_4, g_6\}, \quad X_{x'_1} = \{g_1, g_7, g_9\}, \quad X_{x_2} = \{g_5, g_8\},$$
$$X_{x'_2} = \{g_3\}, \quad X_{o_1} = \{g_2\}, \quad X_{o_2} = \{g_0\}.$$

Directly: the only bundle of three goods, $X_{x'_1}$, is worth 3 to o_2 (only g_1), which holds g_0, worth 4, and 2 to x'_2 (only g_9), which holds g_3, worth 4; after removing one good, a bundle of two goods is worth at most 3 to any other agent, and every agent holds at least 4.

C How Large Must a Ring Be?

Call a *short move* a chain or a ring with at most one pair arrow. A natural simpler algorithm would use only short moves. The next proposition shows that the instance of Appendix B is as small as possible in its number of agents among states where short moves are not enough.

Proposition 1 (when short moves are not enough). *In the core case, let Y be a valid state in which not every agent holds its pair, no free agent can finish, and no short move applies. Then $|F| \geq 2$, $n \geq 6$ and $m \geq 8$; if $|F| \geq 3$, then $n \geq 9$ and $m \geq 12$. The bounds $n = 6$ and $m = 8$ are attained.*

Proof. No chain applies, so (NC) holds, and by Lemma 7 every free agent o holds a good and has $|H_o| \geq |F|$, hence at least $|F|$ blockers. The blockers of two different free agents o, o' are different: a common blocker would have the pair $\{y_o, y_{o'}\}$, with no leftover good, against Lemma 7(b). There is no ring of want arrows, so a walk that repeatedly moves to an agent that wants the current agent's good never repeats an agent; it ends at an agent whose good nobody wants, or that holds nothing, which is free. Every agent holds a good: pair holders two, free agents one by Lemma 7, others a wanted good. Let $k = |F|$; $k \geq 1$, because the walk from an agent outside U ends at a free agent.

If $k = 1$, let $F = \{o\}$ and x a blocker of o. Then x is not free, and its walk ends at o; with the pair arrow $o \to x$ this is a ring with one pair arrow, a short move. So $k \geq 2$.

If $k = 2$, let $F = \{o_1, o_2\}$. Agent o_1 has at least two blockers and only o_2 is another free agent, so some blocker x of o_1 is not free. Its walk cannot end at o_1, as above, so it ends at o_2, entering it by a want arrow ($x \neq o_2$). So o_2 wants a good and does not hold its top: o_2 is not a blocker of o_1, and no blocker of o_1 is free. The same holds for o_2. So $n \geq 2 + 2 + 2 = 6$, and $m \geq n + |J| \geq 6 + 2 = 8$.

If $k \geq 3$, the blockers of the k free agents are at least k^2 different agents, at most k of them free, so $n \geq |F| + (k^2 - k) = k^2 \geq 9$, and $m \geq n + |J| \geq k^2 + k \geq 12$.

Attained. The instance of Appendix B has $n = 6$ ($m = 10$). For $m = 8$: o_1 ranks $g_4 \succ g_5 \succ g_0$, o_2 ranks $g_6 \succ g_7 \succ g_1$, x_1 ranks $g_4 \succ g_1 \succ g_2$, x_1' ranks $g_5 \succ g_3 \succ g_1$, x_2 ranks $g_6 \succ g_0 \succ g_2$, x_2' ranks $g_7 \succ g_3 \succ g_0$, with o_1 holding g_0, o_2 holding g_1 and every x its top. Then $J = \{g_2, g_3\}$, $F = \{o_1, o_2\}$, the blockers of o_2 are x_1, x_1' (leftover goods g_2, g_3) and those of o_1 are x_2, x_2' (leftover goods g_2, g_3); nobody can finish, and no short move applies. □

In the second instance the distinct leftover goods matter: of the four cycles $o_2 \to x \to o_1 \to x' \to o_2$ with two pair arrows, two use the same leftover good twice, and trading along them would give it to two agents; the other two are rings.

No bounded number of pair arrows suffices. For every $k \geq 2$ and $d \geq 1$ there are valid states with k free agents $o_0, \ldots, o_{k-1}$ in which each o_i is the root of a binary tree of depth d whose agents want the goods of their two children, and the leaves of the tree of o_{i+1} hold their tops and are blockers of o_i (indices modulo k), with leftover goods that are distinct within each tree: one of its own per leaf, or the goods of a common pool of k, the leaves of each tree taking them in turn. Every cycle of want and pair arrows there has exactly k pair arrows, and no free agent can finish if and only if $k \leq 2^d$, the number of leaves of a tree. For $k = 2$ and $d = 1$ this is the instance of Appendix B. We omit the proofs of these two facts; both are machine-checked in Lean for every k and d.

D Limits of the Shape

Could every instance have an EFX_0 allocation with all bundles of at most two goods, or with one bundle of exactly three? No. Write $\theta_i(B)$, the *threat* of B to i, for the largest value of $v_i(B \setminus \{h\})$ over $h \in B$ (0 if $B = \emptyset$): agent i does not envy X_j in the sense of EFX_0 if and only if $v_i(X_i) \geq \theta_i(X_j)$, and i is *safe* if this holds for every $j \neq i$. By additivity, $\theta_i(B) = 0$ if $|B| \leq 1$; $\theta_i(B) \leq v_i(B \cap R_i)$, with equality if B has a good outside R_i; and $\theta_i(\{g, h\}) = \max(v_i(g), v_i(h))$.

Lemma 8 (cases of safety). *Let agent i have $R_i = \{a, b, c\}$ with $v_i(a) > v_i(b) > v_i(c) > 0$ and $v_i(a) < v_i(b) + v_i(c)$, and let X be an allocation. Then i is safe in X if and only if one of the following holds:*

(T) *i holds a, and either i also holds b or c, or b and c are not together in a bundle of at least three goods;*
(BC) *i holds b and c;*
(B) *i holds b, and a is alone;*
(C) *i holds c, and a and b are alone;*
(E) *a, b and c are all alone.*

Proof. Write a, b, c also for the values. The sums of subsets of $\{a, b, c\}$ are ordered as $0 < c < b < a < b + c < a + c < a + b < a + b + c$. Let $S = X_i \cap R_i$.

- $S \supseteq \{a, b\}$ or $S \supseteq \{a, c\}$: every other bundle meets R_i in at most one good, so its threat is at most $b < a + c$. Safe: (T).
- $S = \{b, c\}$: every threat is at most $a < b + c$. Safe: (BC).
- $S = \{a\}$: if b, c are in different bundles, each threat is at most $b < a$; if $\{b, c\}$ is a bundle, its threat is $b < a$; if a bundle strictly contains $\{b, c\}$, it has a third good, outside R_i, and its threat is $b + c > a$. So: (T).
- $S = \{b\}$: if a is not alone, its bundle has threat at least $a > b$; if a is alone, every threat is at most $c < b$. So: (B).
- $S = \{c\}$: as before a must be alone, and if b is not alone its bundle has threat at least $b > c$. So: (C).
- $S = \emptyset$: a bundle with a good of R_i and at least two goods has a positive threat. So: (E).

Conversely, each of (T), (BC), (B), (C), (E) describes safe situations above: (T) is the first and third items; (BC) is $S = \{b, c\}$ or $S = \{a, b, c\}$; (B) forces $a \notin S$, so $S = \{b\}$ with a alone, or $S = \{b, c\}$; (C) forces $S = \{c\}$ with a and b alone; and (E) means that S is empty or a single alone good, which is (T), (B) or (C). □

Proposition 2 (limits of the shape). *Take values with $v_i(a_i) > v_i(b_i) > v_i(c_i) > 0$ and $v_i(a_i) < v_i(b_i) + v_i(c_i)$ for every agent.*

(a) Let agents 0, 1, 2 rank $g_0 \succ g_1 \succ p_i$, where p_i is valued only by agent i. No EFX$_0$ allocation has all bundles of at most two goods, and $\{g_0\}, \{g_1\}, \{p_0, p_1, p_2\}$, to the three agents in any order, is EFX$_0$.

(b) Let agents 0, 1, 2, 3 rank $g_0 \succ g_2 \succ p_0$, $g_0 \succ g_2 \succ p_1$, $g_1 \succ g_2 \succ p_2$ and $g_1 \succ g_2 \succ p_3$, with p_i valued only by agent i. Every EFX$_0$ allocation has bundle sizes $4, 1, 1, 1$, and one exists.

Proof. (a) Five goods in three bundles of at most two goods have sizes $2, 2, 1$, so exactly one good is alone. Two agents do not hold their top g_0: case (T) fails for them, and (C) and (E) need two or three alone goods. So both are in case (BC) or (B), and both hold g_1, which is impossible. In $\{g_0\}, \{g_1\}, \{p_0, p_1, p_2\}$ the three agents are in cases (T), (B) and (C).

(b) Each of the tops g_0, g_1 has an agent ranking it first that does not hold it, a *loser*; the two losers differ. A loser is safe only by (BC) or (B), which need it to hold g_2, or by (C) or (E), which need g_2 alone. The two losers cannot both hold g_2, so g_2 is alone, and nobody is in case (BC). A loser holding $\{g_2\}$ needs its top alone; a loser not holding g_2 needs its top and g_2 alone. So g_0, g_1, g_2 are alone, held by three agents, and the fourth bundle is $\{p_0, p_1, p_2, p_3\}$. Such an allocation exists: g_0 to agent 0,

g_1 to agent 2, g_2 to agent 1, the four goods p_i to agent 3 (cases T, B, T, C). □

In (a) every agent values exactly three goods and is strictly balanced, so the one large bundle of Theorem 1 cannot be avoided; by (b), its size cannot be bounded by three.

E Related Work in a Table

An instance that no earlier result covers. Six agents value $\{g_0, g_1, g_6\}$, $\{g_0, g_1, g_7\}$, $\{g_0, g_2, g_8\}$, $\{g_2, g_3, g_4\}$, $\{g_3, g_5, g_9\}$ and $\{g_4, g_5, g_{10}\}$, with pairwise distinct values between 1 and 2; so $n = 6$ and $m = 11$, and every agent values exactly three goods and is strictly balanced. Theorem 1 applies. Row by row, the other results of Table 1 do not:

- $n = 3$ [5] and $n = 4$ [2]: here $n = 6$.
- At most three distinct valuations [7]: the six agents value different sets of goods, so the six valuations differ.
- $m \leq n + 3$ [12] and $m = n + 4$ [4]: here $m = n + 5$.
- At most two agents valuing each good [6,1,13,8] (a good is an edge between the agents that value it): three agents value g_0.
- Hypergraphs of girth at least four [9]: the first two agents both value g_0 and g_1, a cycle of length two.
- At most four relevant goods per agent [14]: covers the instance, but for EFX, not EFX_0.
- At most two relevant goods per agent [11]: here three.

The script of Appendix F checks these properties and runs DE on 2,000 random choices of the values.

Table 1. Existence results for complete allocations under restrictions. *Version cited*: the published version where one exists, else arXiv. [a]Goods nobody values go to an arbitrary agent. [b]Includes additive. [c]Extended abstract with a proof sketch. [d]An AAMAS 2025 extended abstract has the degree and girth conditions. [e]Accepted at EC 2026.

Result	Restriction (agents, goods, who values what)	Notion	Complete	Valuations	Algorithm	Version cited
[5]	$n = 3$	EFX_0	yes	additive	pseudo-polynomial	JACM 2024
[7]	at most three distinct valuations	EFX_0	yes	additive	no bound stated	EC 2025
[12]	$m \leq n + 3$	EFX_0	yes	monotone	none stated	MOR 2024
[2]	$n = 4$, $m \leq 9$	EFX_0	yes	additive	none stated	arXiv
[4]	$m = n + 4$	EFX_0	yes, claimed	additive	none stated	Zenodo, not peer-reviewed
[6]	graphs: at most two valuers per good, two agents share at most one good	EFX_0	yes	monotone	polynomial time	MOR 2026 (EC 2023)
[1]	multigraphs: at most two valuers per good	EFX_0	yes	cancelable[b]	polynomial time	arXiv[e]
[13]	multigraphs: bipartite, or at most $\lceil n/4 \rceil - 1$ neighbours per agent, or girth at least six (non-parallel edges)	EFX_0	yes	monotone	not efficient (stated)	arXiv[d]
[8]	at most two valuers per good, $v_i(g) \in \{0, v_g\}$	EFX_0	yes	restricted additive	no time bound stated	WINE 2024
[9]	hypergraphs of girth at least four (a good joins its valuers)	EFX_0	yes	monotone	polynomial time	AAAI 2026
[14]	at most four relevant goods per agent	EFX	yes[a]	additive	polynomial time, announced	AAMAS 2024[c]
[11]	at most two relevant goods per agent	EFX_0	yes	additive; monotone	serial dictatorship; Lean-checked	GitHub, v1.2.0
this paper	at most three relevant goods per agent; any number of valuers per good	EFX_0	yes	additive	polynomial time, at most $4n$ trades	this paper

F Verification in Detail

Lean. The proofs of Sections 2–6 and of the appendices are machine-checked in core Lean 4 (`lean/EFX/K3DE*.lean` in [10], and `K3Extras.lean` for serial dictatorship): no library beyond Lean's own, no unproved step, every declaration on Lean's three standard axioms, checked by the repository's `lean/check.sh`, which also replays the library in a fresh kernel. DE is a Lean function. To trust the main theorem, one reads the model (`lean/EFX/Model.lean`, about 50 lines: instances with natural-number values, allocations as maps from goods to agents, EFX_0, the number of relevant goods) and the statement of `EFX.DE.de_correct`; Table 2 lists the theorem behind every result. The Lean files use other names for some notions: *needs* for wants, *junk* for leftover goods, *utility* for score, *exposed* for blocker, *protecting good* for the leftover good h_x, *need arcs* and *exposure arcs* for want and pair arrows, *need cycle* and *exchange cycle* for rings, *pair chain* for a chain, (P) for (NC), and *valid absorber* for an agent that can finish (a notion that also allows a pair holder to finish, as when every agent holds its pair). One Lean lemma covers chains and rings (a chain is closed by an arrow from its last agent back to x).

Real values. The model has natural-number values, and real values reach it through an encoding. Let agent i have relevant goods of values $x_1, x_2, x_3 > 0$. (With one or two relevant goods, the missing slots repeat the first good in these comparisons, so some compared sums count a good twice; no set of goods contains a missing slot.) After cancelling the goods common to both sides, a comparison of two sums of its values compares two disjoint sets of these goods; since the values are positive, it is decided by twelve basic comparisons: $x_k \le x_l$ for $k \ne l$, and $x_k \le x_l + x_r$ and $x_l + x_r \le x_k$ for $\{k, l, r\} = \{1, 2, 3\}$. Both directions are asked, so ties are recorded. A program asks an oracle for $\le$ these twelve questions, after m questions $v_i(g) \le 0$ that find the relevant goods, and gives the three goods the first of 31 triples of natural numbers with the same twelve answers (that the table covers every consistent pattern is checked by Lean's kernel); every other good keeps the value 0. The new values w answer every comparison of two sums of an agent's values as v does (`EFX.K3.agree_surrogate`). That suffices: the EFX_0 condition $v_i(X_i) \ge v_i(X_j \setminus \{h\})$ compares two such sums, so an allocation is EFX_0 for w if and only if it is for v, and the relevant goods are the same (`EFX.efx0_iff_of_agree`, `EFX.numRelevant_eq_of_agree`). So DE run

Table 2. The results of this paper and their Lean theorems in [10], in the namespace `EFX.DE` unless the name starts with `EFX.`; each has a `#print axioms` certificate in its file. *Rankings*: stated for every profile of rankings, over DE's states. $\mathbb{N}$: natural-number values, those of the model.

Result	Lean theorem	Values; what is proved
Theorems 1 and 5; Theorem 2 without the time	`de_correct`	$\mathbb{N}$: DE's output is EFX_0, all bundles but at most one have at most two goods, at most n peeling rounds and $4n$ trades
The same, real values	`deOrd_correct`, `thmD_ordered`	any linearly ordered cancellative commutative monoid, such as $\mathbb{R}_{\geq 0}$; nonnegative values
Running time (Theorem 2)	`de_eq_spec`, `deC_cost`	$\mathbb{N}$: a counted program returns DE's output within $750(n+1)(n+m+1)$ units
The same, real values	`deOrdC_cost`, `EFX.K3.surrogateC_cost`	as above, plus the encoding: at most $n(m+12)$ comparisons and $10nm + 971n$ other units
The encoding of real values	`EFX.K3.agree_surrogate`, `EFX.efx0_iff_of_agree`	ordered values
Lemmas 1 and 2; R1 for two goods	`peeling`, `R1fail`, `EFX.Inst.serialDict_efx0`	$\mathbb{N}$
Lemmas 3 and 4	`draft_valid`, `transfer`	rankings
Theorem 3	`soundness`	$\mathbb{N}$; every valuation consistent with the rankings (balance $v_i(a_i) \leq v_i(b_i) + v_i(c_i)$ suffices)
Lemmas 5 and 6	`exchange`, `exchange_scores`	rankings; one lemma for both
Lemma 7	`absorber_iff`, `not_exposed_of_none`	rankings
Theorem 4, Corollary 1	`improvement`, `completable_of_undominated`	rankings
Algorithm 1, the loop	`loop_spec`, `reps_first_blocker`, `step_next_scores`	rankings
The examples of Sections 1 and 6, Appendix B	`Remarks.efx_not_efx0_intro`, `Examples.small_spec`, `Examples.worked_spec`, `ShortExamples.w_exactly_four`, `ShortExamples.w_completions`, `Remarks.w_cycles`	by evaluation
Proposition 1; the ring family	`prop_short`, `ShortExamples.w_attains`, `ShortExamples.s8_attains`, `Rings.nExp_eq`, `Rings.no_free_absorber_iff`	rankings; every k and d
Lemma 8, Proposition 2	`Limits.safe_iff_cases`, `Limits.limits_a`, `Limits.limits_b`	$\mathbb{N}$, every value profile of the stated form; also ordered values

on w returns an allocation that is EFX_0 for v, of the same shape and within the same bounds (`EFX.DE.deOrd_correct`).

Running time. A counted program, proved to return exactly DE's output (`de_eq_spec`), takes at most $750(n+1)(n+m+1)$ units, reading the input included (`deC_cost`), with one unit per elementary operation, comparison, array read and array write, as in a random-access machine, arrays being used single-threaded; a natural number costs one unit whatever its size. On real values the encoding adds at most $n(m+12)$ comparisons and $10nm+971n$ other units (`EFX.K3.surrogateC_cost`, `deOrdC_cost`).

What is not formalized. (1) That the units match the steps of a machine: a property of the counted program, checked by reading it. (2) That the nonnegative reals satisfy the axioms of Lean's class of values: core Lean has no real numbers, so this textbook fact is assumed (the instances for natural numbers and integers are checked). (3) That DE run directly on real values makes the same decisions as on the encoded values: true because every decision is a comparison that the encoding preserves, but Lean states the theorem for DE run on w. (4) That the Lean statements say what the text says. They were written by AI agent sessions of the coding assistant that helped with this work (Claude Code) and compared with the text by independent AI sessions (`k3/simplify/po/referee/lean_audit*.md` in [10]), not by a human; the model and the statement of `de_correct` are short enough to check by hand. (5) The second proof mentioned below, the novelty claim, and the tests.

Tests. The tests are evidence, not proof. Three implementations, each asserting every claim of a proof and checking outputs against the raw definition of EFX_0 with exact arithmetic, ran without failure on every profile with $n = 2$ and $m \le 6$, with $n = 3$ and $m \le 7$, and with $n = 4$ and $m \le 6$ (in part), on cores with $n = 5, 6$ and on random profiles with up to 9 agents: about 1.7 million valid states for the proof of Section 5, about 15.5 million for an independently found second proof (a walk instead of a choice of distinct leftover goods), and about 2.9 million runs of a referee's implementation written from the text alone (`k3/simplify/po/` in [10]). The first and the referee's run DE's loop with the same three steps (up to the choice among leftover goods); the second tests the lemmas of Sections 4 and 5 on its own loop, which lets a free agent holding nothing finish first. The script `paper/k3-simple/examples/check_examples.py`, written from this paper with the loop as stated here, recomputes every exam-

ple and number of the paper and runs DE, peeling included, on random instances with ties, unbalanced agents and goods nobody values.